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Net contributions of multiple oceanic warm and cold thermal events to tropical cyclone intensity revealed by idealized coupled simulations

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Abstract
The ocean serves as the energy source for tropical cyclones (TCs), and a TC typically encounters multiple oceanic warm and cold events during its lifecycle. While most existing studies emphasize individual thermal events, the net contribution of multiple warm and cold events to TC intensity remains unclear. We use a set of idealized coupled atmosphere–ocean simulations to examine how multiple ocean warm and cold thermal events affect TC intensity evolution throughout its lifecycle. Based on observational evidence, our experimental design includes one control experiment and 30 sensitivity experiments across three core scenarios (warm-anomaly dominated, cold-anomaly dominated, and warm-cold balanced anomalies). Relative to the control experiment, warm dominated experiments produce a stronger intensity evolution, cold dominated experiments produce a weaker decay, and the balanced experiments also lead to a slight increase. This response is fundamentally asymmetric, as the intensifying influence of warm events outweighs the weakening influence exerted by cold events with comparable magnitude. Moreover, this asymmetry increases with the magnitude of the oceanic thermal anomalies, and when the warm-anomaly amplitude is doubled, the simulated TC satisfies the rapid intensification criterion. These findings quantify the asymmetric forcing of ocean thermal structures on TCs, providing a crucial mechanistic basis for improving TC intensity forecasts.

Keywords Tropical cyclone, Air–sea coupling, Latent heat flux, Mesoscale thermal events, Upper-ocean heat content

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Introduction

Tropical cyclones (TCs) derive their primary energy from the ocean via air–sea enthalpy fluxes (Emanuel 1986; Wadler et al. 2023). While high sea surface temperatures (SST) are a prerequisite for genesis (Emanuel 1988; Gilford et al., 2021; Done et al. 2022), the subsequent intensity evolution is heavily regulated by a negative feedback mechanism: the storm's strong winds induce mixing and upwelling, which cools the sea surface and throttles the energy supply (Price 1981; Cione and Uhlhorn 2003; Paul et al. 2024; Jaimes and Shay 2025). Consequently, the magnitude of this self-induced cooling is dictated by the pre-storm upper-ocean thermal structure. Accurately representing this vertical thermal structure is therefore essential, as atmosphere-only models often overestimate intensity by failing to account for the ocean's dynamic cooling response (Bender and Ginis 2000; Srinivas et al. 2016; Gramer et al. 2024).

Rather than being a spatially uniform background, the real ocean presents a heterogeneous mosaic characterized by ubiquitous warm and cold features, including mesoscale eddies, oceanic fronts, and marine heatwaves (Hobday et al. 2016; Schlegel et al. 2021). As a TC traverses this complex environment, it experiences alternating periods of suppressed and enhanced cooling—warm eddies insulate the storm from negative feedback, while cold wakes accelerate weakening (Jaimes and Shay 2009). Because these thermal features are densely populated across the ocean basin, a TC inevitably interacts with a sequence of successive thermal events throughout its lifecycle, rather than an isolated anomaly. The cumulative impact of these complex interactions poses a significant challenge for understanding intensity changes.

Numerous studies have documented the influence of oceanic warm and cold events on TC intensity. These thermal contrasts can control the duration and magnitude of surface enthalpy supply. Notable cases illustrate this sensitivity (Fig. 1). For instance, Supertyphoon Maemi (2003) intensified rapidly while crossing a prominent warm eddy in the western North Pacific, consistent with the idea that a thicker warm layer can suppress SST cooling and support larger enthalpy flux into the storm (Lin et al. 2005; Wu et al. 2007). Furthermore, recent work indicates that marine heatwaves can

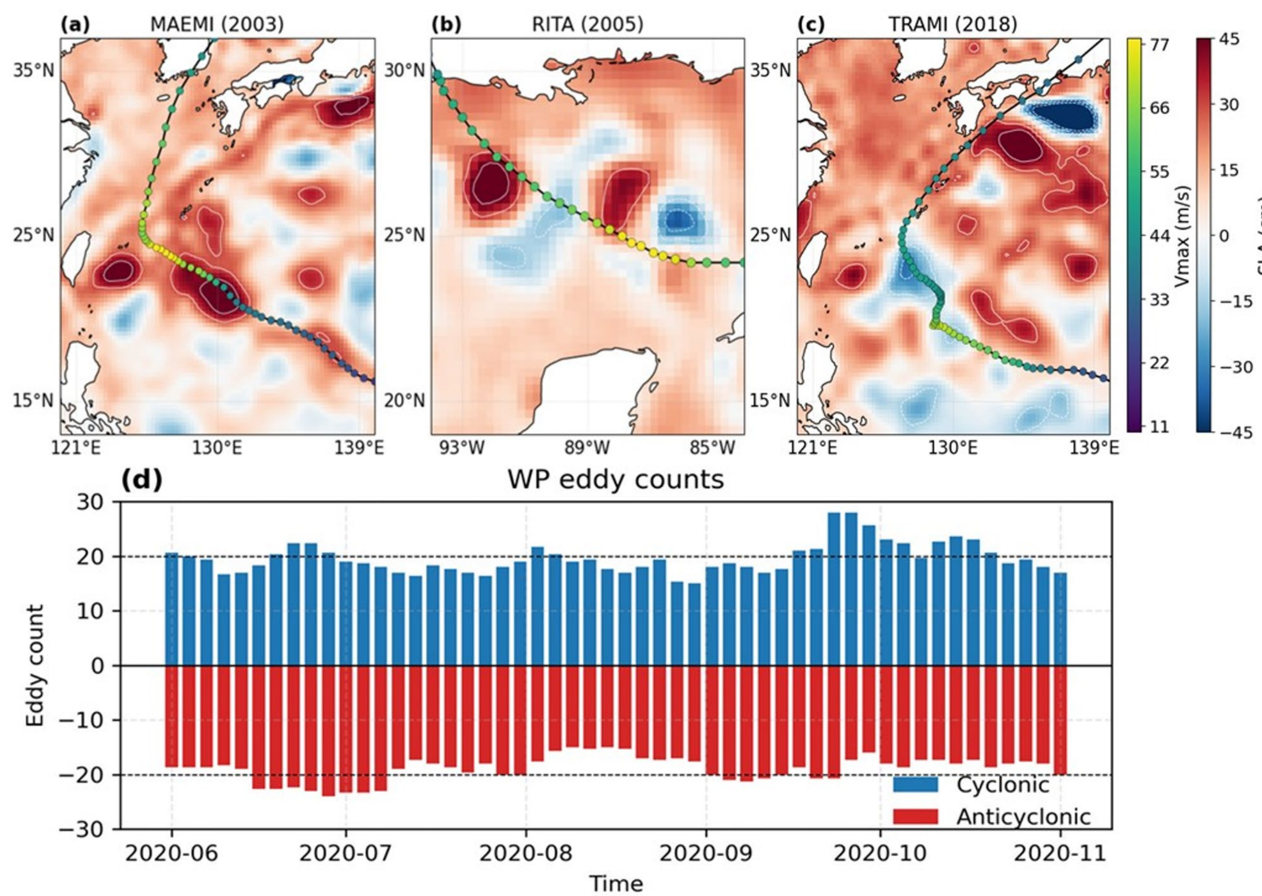

Fig. 1 Statistics of real tropical cyclone (TC) case studies and ocean eddy occurrences. Panels (**a**–**c**) show the intensity evolution of Maemi (2003; Lin et al. 2005), Rita (2005 ; Jaimes and Shay 2009), Trami (2018; Wada 2021), respectively, as they encounter ocean eddies, with the background shading representing sea level anomaly (SLA). Panel (**d**) shows the time series of the number of mesoscale ocean eddies in the western North Pacific

enhance tropical cyclone intensity through increased latent heat flux near the TC core, highlighting the importance of event-scale thermal anomalies in setting air–sea energy supply; more generally, elevated upper-ocean heat content and a deeper warm layer can form part of the favorable oceanic background for TC rapid intensification (Paul et al. 2022; Choi et al. 2024; Radfar et al. 2024). Conversely, Cold features can have the opposite effect. Observations for Hurricanes Katrina and Rita (2005) show enhanced mixed layer cooling and weakening after passage over cold core eddies, highlighting how a reduced ocean heat reservoir can inhibit intensification (Jaimes and Shay 2009). Rapid weakening linked to eddy rich conditions has also been documented in high resolution coupled simulations for typhoons such as Trami and Kong-Rey (2018; Wada 2021).

Despite this progress, most mechanistic understanding is still derived from idealized settings that isolate a single warm or cold feature, or from composites conditioned on one event type. Sun et al. (2023) found in their study of Typhoon Hato (2017) that an anticyclonic eddy near the TC core tends to enhance intensification, whereas anticyclonic eddies in the outer region can weaken the TC by diverting low-level inflow, implying that the net effect of grouped eddies is strongly position-dependent and remains difficult to generalize. Real TCs commonly traverse a patchy field with both warm and cold anomalies, Ma et al. (2017) reported that over 90% of TCs in the western North Pacific encounter mesoscale eddies, meaning a storm typically samples a sequence of multiple features within its lifecycle. For complex oceanic conditions, Kumar et al. (2026) found in a one-year storm- and eddy-resolving global coupled simulation that TCs spending more time over cold-core than warm-core eddies tend to show lower mean intensity and intensification rates. Consequently, a critical question remains unanswered: it is unknown whether the enhancing effects of warm events and the suppressing effects of cold events linearly cancel each other out, or if the storm responds asymmetrically to this complex, alternating thermal forcing.

Importantly, the distribution of the number of warm and cold eddies in the ocean often exhibits an asymmetrical phenomenon (Fig. 1d). This motivates examining whether warm and cold anomalies of the same amplitude produce an asymmetric net impact on western North Pacific TC intensity. Guided by observational evidence, we perform a total of 31 coupled simulations, comprising one control experiment and 30 sensitivity experiments. These experiments impose warm-anomaly dominated, cold-anomaly dominated, and warm-cold balanced ocean events as patchy temperature anomalies in the initial ocean state, with the atmospheric environment and TC initialization held fixed. The experiments show that ocean thermal patchiness alters TC intensity compared to the control, producing an asymmetric response in the western North Pacific, such that warm events enhance TC intensity more strongly than cold events suppress it. Moreover, this asymmetry increases with the magnitude of the oceanic thermal anomalies.

Model configuration and experiment design

Model configuration

Given that TC–ocean interaction is not one-way, but includes both oceanic modulation of TC intensity and TC-induced ocean responses that can extend to mesoscale cyclonic eddy generation (Lin et al. 2025), a two-way air–sea coupled modeling framework is necessary. In this study, we use the COAWST framework, which integrates the Weather Research and Forecasting (WRF, version 4.5; Skamarock et al. 2019) model and the Regional Ocean Modeling System (ROMS; Shchepetkin and McWilliams 2005) via the Model Coupling Toolkit (Larson et al. 2005; Warner et al. 2010).

The atmospheric (WRF) and oceanic (ROMS) components exchange surface momentum, heat fluxes, and SST every 900 s. The simulation domain spans from 16°N to 33°N and 115°E to 134°E. Both models have a horizontal resolution of 4.5 km, which is sufficient to resolve mesoscale and partially submesoscale processes in both the atmosphere and ocean. In ROMS, vertical turbulent mixing is parameterized using the generic length scale (GLS) scheme (Warner et al. 2005; Liu et al. 2024).

An idealized, axisymmetric warm-core vortex (Rotunno and Emanuel 1987) is embedded into a horizontally uniform background atmosphere based on the Jordan (1958) tropical sounding (Xiong and Wu 2025). The initial vortex has a maximum 10-m wind speed of approximately 29 m s^{-1} and a minimum surface pressure of about 1002.1 hPa, is illustrated in Fig. 2a. The corresponding perturbation fields are then interpolated onto the August climatological background field from ERA5 (Hersbach et al. 2023) and superimposed directly within a 300 km radius of the TC center. To guide the typhoon along a consistent northward track across all experiments, a uniform steering flow of $(\mathrm{U}, \mathrm{V}) = (2{,}10)\ \mathrm{m\,s^{-1}}$is imposed within a 300 km radius of the initial center (Wu and Chen 2016). The initial conditions for ocean temperature and salinity are derived from the GLORYS12V1 global ocean reanalysis product (Copernicus Marine Service product identifier: GLOBAL_MULTIYEAR_PHY_001_030).

Experimental design

To represent oceanic warm-cold anomalies, we employ random eddy-like anomaly fields, which are conveniently described mathematically. According to the eddy dataset (Fig. 1d), the numbers of cold and warm eddies within a given region are generally unbalanced. Therefore, we

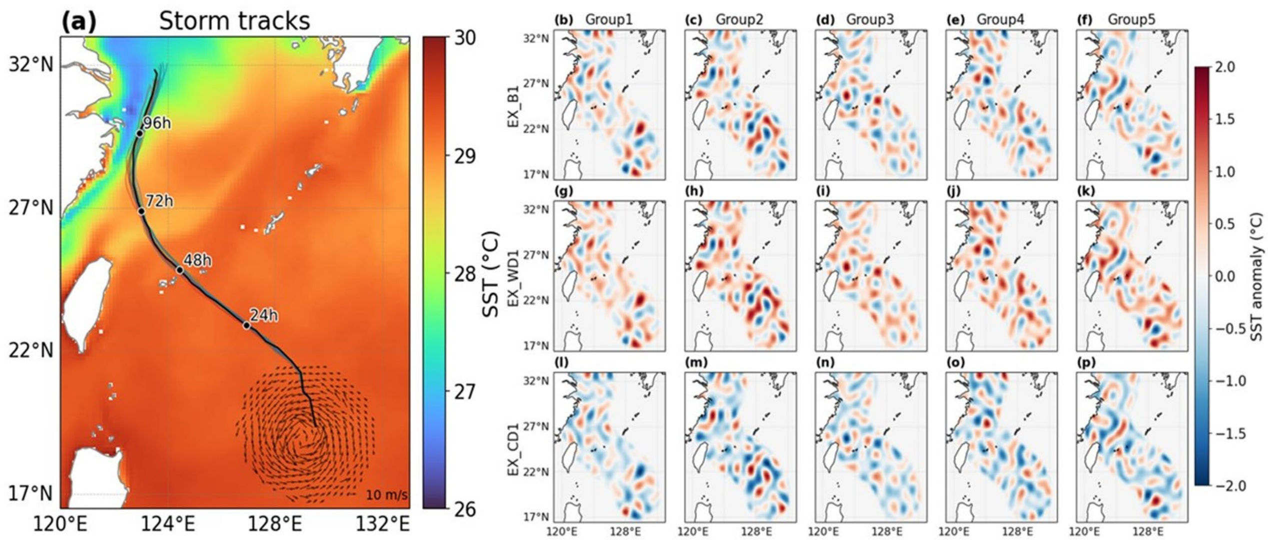

Fig. 2 Model configuration and the control experiment. In (**a**), the arrows show the embedded 10 m wind field, illustrating the structure of the idealized TC, and the shading denotes the climatological SST in the control experiment. Thin solid lines indicate TC tracks for all experiments, with the black solid line representing the control track. Panels (**b**–**p**) show the SST anomalies in each experiment relative to the control

consider three scenarios: warm-anomaly dominated, cold-anomaly dominated, and warm-cold balanced anomalies.

We first conduct a set of control experiments in which the ocean state is taken from the August climatology of the GLORYS12V1 product, which provides the translation track of the TC (Fig. 2a). In the subsequent experiments, warm-cold anomalies are superimposed onto the August climatological ocean state within 300 km of the TC track. The random warm and cold anomalies are described by

$$\xi\left(x,y,z\right)=\sum k_x,k_y\exp\left[-\frac{\left(k-\omega_n\right)^2}{2\sigma_k^2}\right]\sin\left(2\pi k_x\frac{x}{L_x}+\theta_1\right)\sin\left(2\pi k_y\frac{y}{L_y}+\theta_2\right)I\left(z\right), \quad (1)$$

where x and y denote the discrete grid indices, z denotes depth (positive downward), k_x and k_y are integer wavenumber indices in the zonal and meridional directions, L_x and L_y are the domain lengths, and $\theta_1, \theta_2 \in [0,2\pi)$ are independent random phases. Here $k=\sqrt{k_x^2+k_y^2}$, $\omega_n=7.8$, and $\sigma_k=3.5$.

The warm dominated and cold dominated anomalies are given by

$$\Delta T_{\mathrm{warm}}=AN\left(\xi+bI\right),\Delta T_{\mathrm{cold}}=-\Delta T_{\mathrm{warm}}, \quad (2)$$

and the warm-cold balanced anomaly is constructed as

Table 1 Summary of experimental configurations and their upper-ocean heat content in the upper 100 m (UOHC100) anomalies (10^7 J m^{-2})

ID	Description	Objective	UOHC100 Groups (Mean)
EX_B1	Warm-cold balanced	Moderate amplitude pattern experiments	0, 0, 0, 0, 0 (0)
EX_WD1	Warm-anomaly dominated		5.55, 7.17, 5.76, 6.39, 7.17 (6.41)
EX_CD1	Cold-anomaly dominated		− 5.55, − 7.17, − 5.76, − 6.39, − 7.17 (− 6.41)
EX_B2	Double amplitude warm-cold balanced	Double amplitude sensitivity experiments for the same patterns	0, 0, 0, 0, 0 (0)
EX_WD2	Double amplitude warm-anomaly dominated		11.1, 14.3, 11.5, 12.8, 14.3 (12.8)
EX_CD2	Double amplitude cold-anomaly dominated		− 11.1, − 14.3, − 11.5, − 12.8, − 14.3 (− 12.8)

$$\Delta T_{\mathrm{bal}}=AN\left(\xi-\frac{\sum\xi A_g}{\sum A_g}\right), \quad (3)$$

where $b=0.1$, A_g is the grid area and $\mathcal{N}(\cdot)$ denotes normalization to [−1,1] over the active ocean area. Here A denotes the maximum amplitude of the imposed sea temperature anomaly. The vertical weighting function is defined as

$$I\left(z\right)=\begin{cases}1, & z\leq 100m,\\ sin\left(\pi\frac{z-100}{1800}+\frac{\pi}{2}\right), & 100m<z<1000m\\ 0, & z\geq 1000m.\end{cases}, \quad (4)$$

In addition to the control run, we conduct six groups of experiments (Table 1). The first three groups correspond to warm-anomaly dominated (EX_WD1), cold-anomaly dominated (EX_CD1), and warm-cold balanced anomalies (EX_B1), in which the imposed SST anomalies have moderate amplitude ($A = 2$℃). The structures of these initial eddy fields are shown in Fig. 2. To examine the relative strength between the TC and the imposed anomalies, the last three groups use double amplitude sea temperature anomalies ($A = 4$℃), denoted as (EX_WD2, EX_CD2, EX_B2). To reduce randomness, each group is repeated five times, resulting in a total of 30 sensitivity experiments.

Metrics for tropical cyclone intensity change, latent heat flux, and upper-ocean heat content

To quantify the cumulative influence of ocean perturbations on TC intensity over the full simulation period, we compute the integrated intensity anomaly, IV, as:

$$IV = \sum_{t=t_0}^{t_1} [V_{\text{case}}(t) - V_{\text{ctrl}}(t)] \Delta t, \tag{5}$$

where $V_{\text{case}}(t)$ and $V_{\text{ctrl}}(t)$ are the TC intensity at time t in the sensitivity experiment and the control experiment. Δt is the output interval (set to 3 h). Positive (negative) IV indicates a net strengthening (weakening) relative to the control when accumulated over the entire period.

The surface latent heat flux (LHF), which represents the turbulent flux of moisture from the ocean to the atmosphere via evaporation, is taken directly from the WRF model output variable LH (units: W m^{-2}).

To quantify ocean warm and cold events in a way that reflects subsurface structure, we compute upper-ocean heat content in the upper 100 m, UOHC100, as:

$$\text{UOHC100} = \rho_0 C_p \int_0^{100\,\text{m}} T(z)\, dz \tag{6}$$

where ρ_0 is a reference seawater density (set to 1025 kg m^{-3}), C_p is the specific heat capacity of seawater (set to 3985 J kg^{-1} K^{-1}), $T(z)$ is the ocean potential temperature, z is geometric depth measured positive downward from the sea surface. The UOHC100 anomalies for each experiment are shown in Table 1. The oceanic UOHC100 change for the balanced ocean events is zero, and the magnitudes of the UOHC100 anomalies for the warm dominated and cold dominated events are comparable.

Results

TC intensity evolution and asymmetric

The simulation spans 120 h, with the TC initiating at 19°N, 129°E and moving northwestward. It reaches the vicinity of the Ryukyu Islands in approximately 48 h (Fig. 2a), ensuring a sufficiently long period over the open ocean for it to interact with the underlying oceanic conditions. The control experiment provides a reference track of TC development, against which the impact of ocean patchiness can be measured. In the control run, the intensity increases steadily and reaches its lifetime maximum intensity (LMI) of 46.2 m s^{-1} at 63 h (Fig. 3).

When warm anomalies dominate (EX_WD1), the TC strengthens faster during the early and middle stages. By 48 h, the intensity exceeds the control by about 4 m s^{-1}, the integrated intensity anomaly of TC (IV; Eq. (7)) relative to the control is 136 m s^{-1} h throughout the lifetime, corresponding to an average intensity increase of approximately 3.7%. Yet, the LMI remains close to the control at 45 m s^{-1}. In contrast, the EX_CD1 shows a persistent

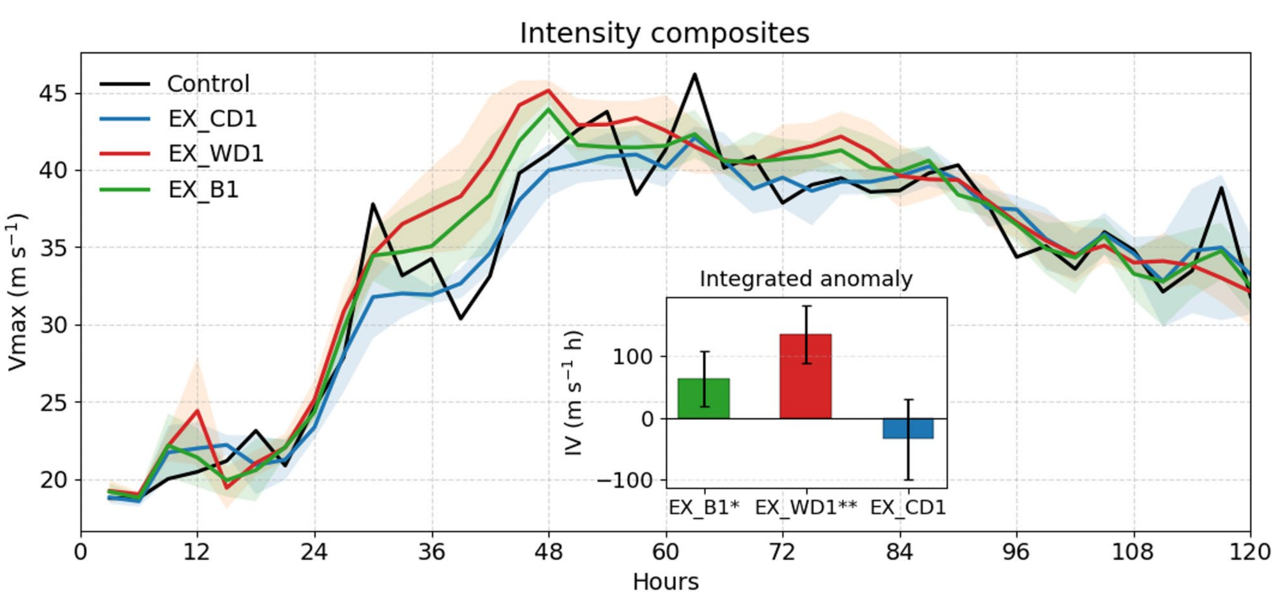

Fig. 3 TC intensity (m s^{-1}) and integrated intensity anomaly (m s^{-1} h). The main panel shows composite time series of TC intensity for each experiment group. The inset shows the mean integrated intensity anomaly for each group relative to the control. Shading indicates one standard deviation. In the inset, * and ** indicate that the difference from the control experiment is significant at the 95% and 99% confidence levels, respectively, based on a t test

deficit throughout most of the life cycle. The TC intensifies more slowly and reaches a lower LMI of 42 m s^{-1}, and this reduction is statistically significant ($p < 0.01$). The integrated intensity anomaly is -34 m s^{-1} h under the same atmospheric constraints, translating to an average intensity reduction of about 0.4%. In the B case, where warm and cold patches are sampled along the track, the integrated intensity anomaly is 63 m s^{-1} h rather than being nearly zero, which represents an average enhancement of roughly 1.8%. This indicates that, although the imposed EX_WD1 and EX_CD1 perturbations are designed to have comparable magnitude, the intensity response is not symmetric. The positive intensity evolution in the EX_WD1 accumulates more over the lifecycle than the intensity reduction observed in the EX_CD1.

Ocean heat content control on latent heat flux and asymmetric intensity response

The upper-ocean heat content in the upper 100 m provides a measure of how the imposed anomalies alter the ocean energy reservoir accessible to TC. In the EX_WD1, the mean initial UOHC100 ahead of TC is above the control by about 6.41×10^7 J m^{-2}, whereas it is -6.41×10^7 J m^{-2} below control in the EX_CD1 (Table 1). The UOHC100 of EX_B1 stays close to the control. Despite this symmetry in UOHC100, the surface energy supply to TC responds asymmetrically. Averaged within a 100 km radius of the TC center, the EX_WD1 sustains a higher mean LHF of about 301 W m^{-2}, roughly 26 W m^{-2} higher than control (275 W m^{-2}), whereas the EX_CD1 exhibits a suppressed mean LHF of 261 W m^{-2}; EX_B1 is close the control value at 278 W m^{-2} (Fig. 4 (a-c)).

Although the EX_WD1 and EX_CD1 impose equal magnitude anomalies in UOHC100, the intensity response is asymmetric because the pathway from ocean

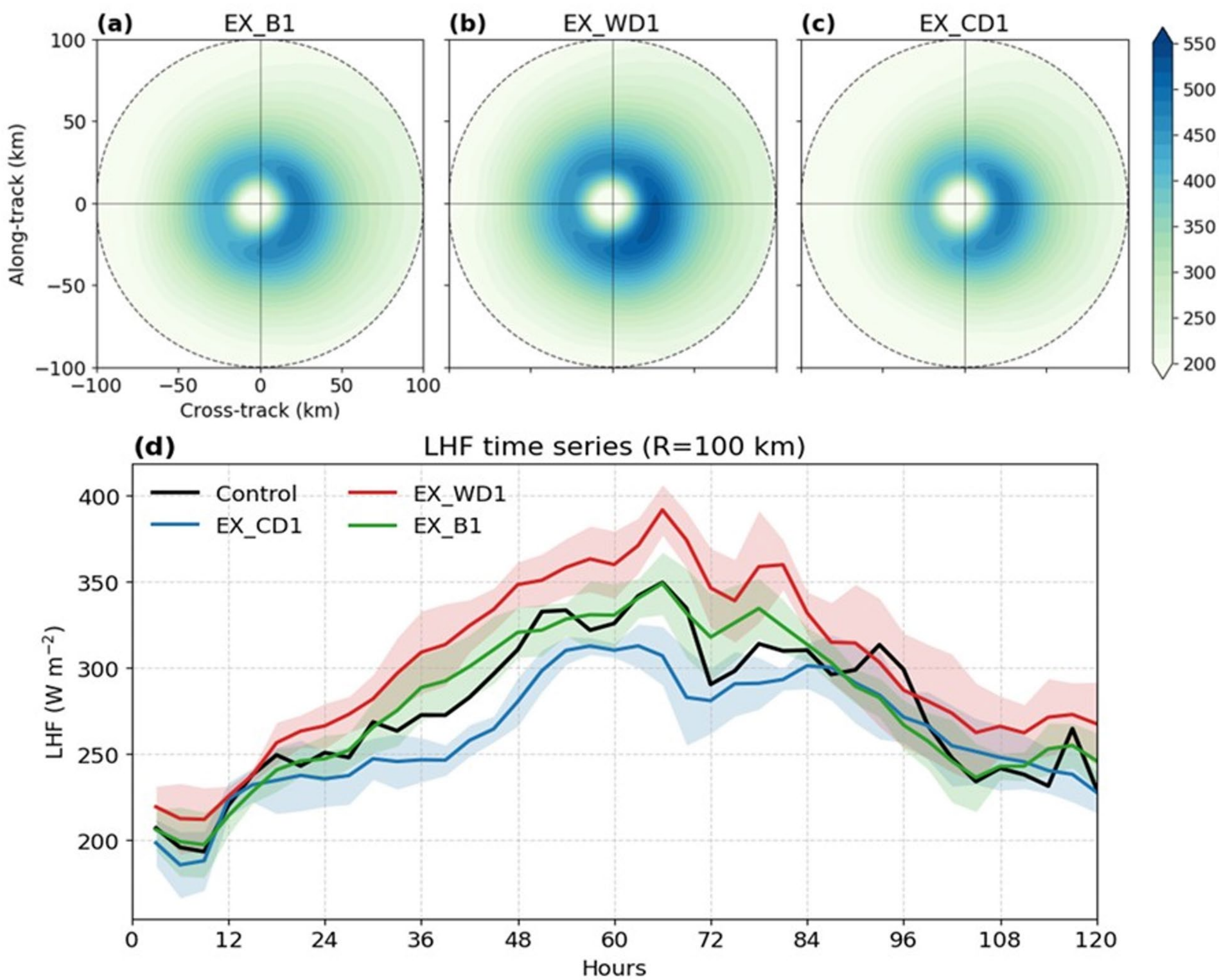

Fig. 4 Composite latent heat flux (LHF). Panels (**a**-**c**) show composite LHF (W m^{-2}) anomaly maps within 100 km of the TC center for the EX_B1, EX_WD1, and EX_CD1, respectively, relative to the control. Panel (**d**) shows the composite time series of LHF for each case, with shading indicating one standard deviation

heat content to TC energy supply is nonlinear (Schade and Emanuel 1999). When encountering a warm eddy, the enhanced UOHC100 increases the effective ocean heat reservoir, suppresses storm-induced cooling caused by TC-driven vertical mixing and upwelling (Mainelli et al. 2008), and thereby sustains higher LHF within 100 km of the TC center (Fig. 4d), a mechanism also supported by coupled and numerical experiments in both the western North Pacific and the Bay of Bengal (Anandh et al. 2020; Ma et al. 2024). The resulting flux surplus persists over a substantial portion of the life cycle, so its influence accumulates and yields a stronger intensity evolution than the control, even though the LMI does not increase. The diagnosed maximum potential intensity (MPI) is higher in EX_WD1 than in the control, consistent with the enhanced enthalpy flux and the corresponding increase in thermodynamic potential (Emanuel 1986; Xu et al. 2019). However, the simulated LMI remains substantially lower than the diagnosed MPI in all experiments, indicating that MPI is not the active limiting factor here. Instead, the additional oceanic energy supply is expressed mainly as stronger cumulative intensification, rather than as a higher peak intensity, under the same large-scale environment and internal TC dynamics. Conversely, when encountering a cold eddy, the reduced UOHC100 favors stronger cooling and lowers LHF, which weakens the TC relative to the control and, importantly, produces a clear reduction in LMI (Lin et al. 2013; Guan et al. 2025). In summary, while the UOHC100 anomalies for warm and cold eddies are symmetric in magnitude, the LHF they deliver to the TC is asymmetric. Consequently, TC intensity is more sensitive to warm eddies than to cold eddies, resulting in stronger intensification under EX_WD1 compared to the weakening under EX_CD1 of comparable magnitude. Furthermore, even in the EX_B1 with offsetting warm and cold anomalies, the net effect on TC intensity remains slightly positive relative to the control.

Asymmetry under amplified thermal anomalies

It is important to note that the primary experiments used a fixed initial TC intensity. To systematically test the sensitivity of TC intensity to the amplitude of oceanic thermal anomalies, a further set of experiments is conducted. When the amplitude of the warm and cold events is doubled, the corresponding intensity responses are shown in Fig. 5a. In the EX_WD2, TC's LMI still does not increase clearly relative to the control, but the integrated intensity anomaly increases markedly to about 400 m s^{-1} h, nearly three times the value in the standard EX_WD1 and corresponding to an average intensity enhancement of 10.2% relative to the control. Notably, under the EX_WD2, the intensity change prior to LMI reaches 67 km h^{-1} over a 24 h period, satisfying the criterion for rapid intensification (RI). This result indicates that patchy warm anomalies are not merely secondary enhancers of TC intensity, but can be sufficiently strong to trigger RI when their amplitude is large. In contrast, doubling the cold event amplitude (EX_CD2) produces little additional change, the LMI remains nearly unchanged and the integrated intensity anomaly is about −23 m s^{-1} h, comparable to −34 m s^{-1} h in the standard cold event case (EX_CD1), which translates to a negligible average intensity reduction of 0.1% relative to the control. Furthermore, under the EX_B2 scenario with equally intensified warm and cold anomalies, TC's integrated intensity anomaly also increases notably, rising from 63 m s^{-1} h in the standard case to 175 m s^{-1} h, representing an average intensity increase of 4.5% relative to the control.

This enhanced asymmetry is consistent with the LHF response (Fig. 5b). Averaged over the storm lifetime within 100 km of the TC center, EX_WD2 sustains a mean LHF of about 349 W m^{-2}, approximately 48 W m^{-2} higher than in EX_WD1, whereas EX_CD2 exhibits a mean LHF of about 260 W m^{-2}, only about 1 W m^{-2} lower than in EX_CD1. In addition, EX_B2 reaches about 297 W m^{-2}, which is about 19 W m^{-2} higher than in EX_B1. These results indicate that, under doubled UOHC100 anomalies, the positive LHF response in the warm-anomaly dominated experiments is amplified much more strongly than the negative LHF response in the cold-anomaly dominated experiments, consistent with stronger positive SST anomalies in the warm cases and weaker negative SST anomalies in the cold cases. This stronger asymmetry in surface energy supply likely explains why the lifecycle integrated intensity anomaly response becomes even more asymmetric when the anomaly amplitude is doubled. These findings further suggest that the amplitude and spatial configuration of ocean thermal anomalies are critical for determining the net TC intensity response.

Summary and discussion

TCs encounter multiple oceanic warm and cold thermal events throughout their lifecycle, yet the net intensity change resulting from these mixed influences is not well quantified. This study examines how such thermal events, represented by patchy warm and cold anomalies imposed in the initial upper-ocean state, shape TC intensity over the full life cycle. Using a series of ocean-atmosphere coupling simulations, we find that ocean patchiness could alters intensity evolution in an asymmetric way. Specifically, when EX_WD1 and EX_CD1 have comparable anomaly magnitudes, the positive intensity anomaly under EX_WD1 is stronger than the negative intensity anomaly under EX_CD1. Notably, the EX_B1 does not yield a neutral intensity response but instead exhibits a net intensification, further supporting that the positive

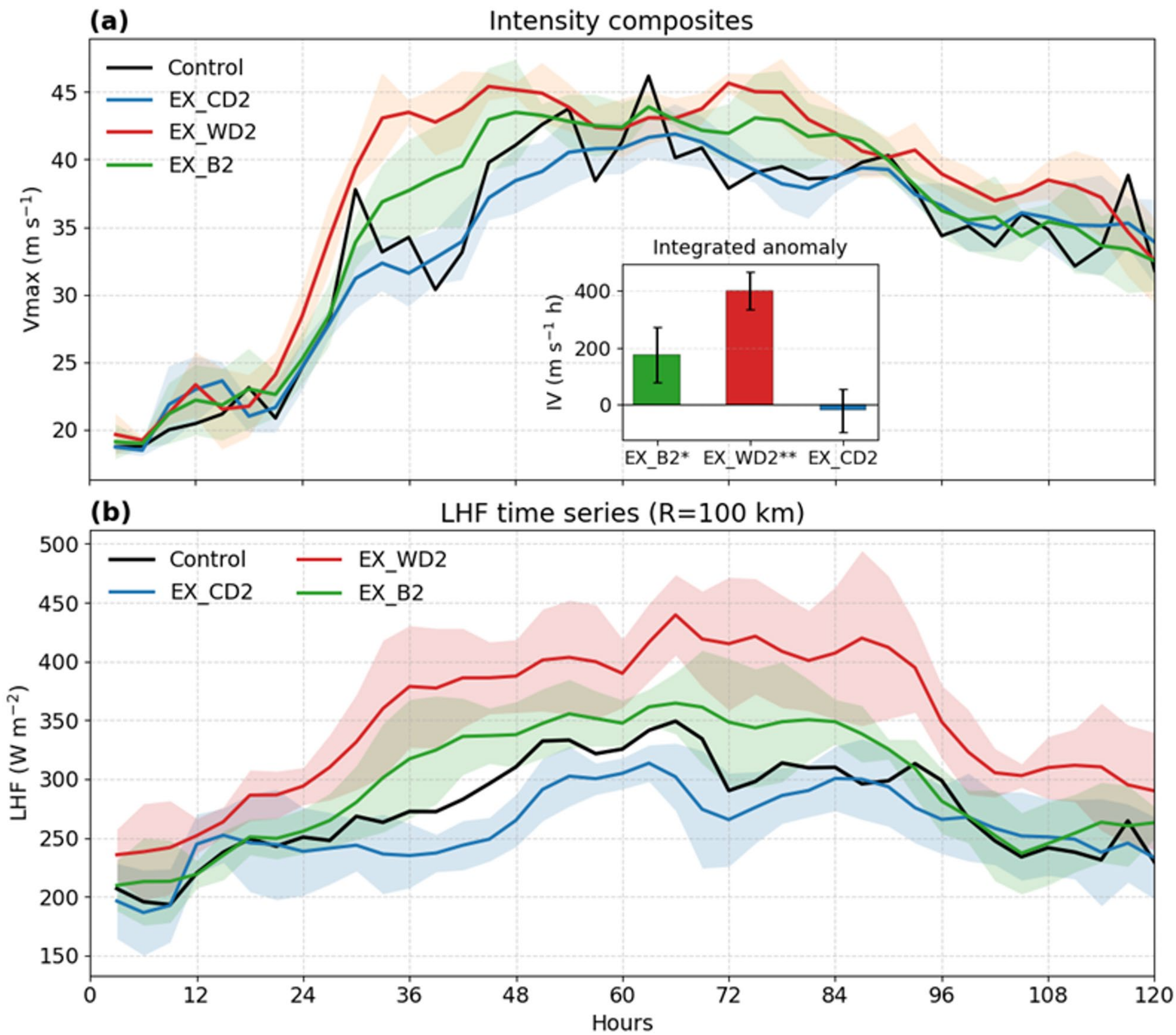

Fig. 5 Intensity composites and LHF composites for experiments with doubled anomaly amplitude. **a** Same as Fig. 3, showing the composite TC intensity (m s^{-1}) and integrated intensity anomaly (m s^{-1} h), but for experiments with doubled anomaly amplitude. **b** Same as Fig. 4d, showing the composite LHF (W m^{-2}) within 100 km of the TC center, but for experiments with doubled anomaly amplitude

contribution from warm eddies outweighs the negative contribution from cold eddies. This asymmetry is physically explained by the differential modulation of LHF, the EX_WD1 exhibits a larger positive LHF anomaly than the magnitude of LHF suppression in the EX_CD1. When the amplitude of the thermal anomalies is doubled, the LHF response is accordingly amplified, leading to a further enhancement of the asymmetry in TC intensity change: the enhancing influence of EX_WD2 becomes stronger, while the suppressing influence of EX_CD2 becomes even weaker. Besides, EX_WD2 satisfies the RI criterion, suggesting that patchy warm anomalies can trigger RI and may otherwise lead to underestimated RI likelihood.

Warm eddies with elevated ocean heat content can favor intensification by limiting TC induced cooling, while cold eddies enhance cooling and suppress strengthening (Mainelli et al. 2008; Lin et al. 2013). However, most prior work has emphasized responses to isolated eddies or a single event class. The real ocean is thermally more complex, and the impact on TC intensity becomes harder to generalize when multiple anomalies coexist and when their position relative to the evolving TC structure changes (Sun et al. 2023). By imposing symmetric magnitude UOHC100 anomalies, our experiments provide a clean attribution framework for understanding why the integrated TC intensity response can differ. Furthermore, we find that when the ocean heat content anomaly is doubled, the enhancing influence of warm-anomaly dominated events increases threefold, whereas the suppressing influence of cold-anomaly dominated events weakens. This occurs because, in the double amplitude sensitivity experiments, the positive contribution of embedded warm eddies to lifecycle integrated intensity increases more strongly than the additional negative contribution from cold eddies, thereby further widening the asymmetry in the net integrated intensity response. This quantified nonlinearity underscores that the net contribution of multiple oceanic warm and cold thermal events to TC intensity is asymmetric and highly sensitive to anomaly amplitude.

Several limitations should be acknowledged. First, the imposed ocean anomalies are idealized and focus on eddy-like thermal structure because it can be introduced cleanly with a simple mathematical formulation. Other oceanic mesoscale processes, such as thermal fronts, marine heat waves and cold spells, may influence TC intensity through similar mechanisms. Second, due to computational constraints, the experiments focus only on a TC with moderate initial intensity, the sensitivity to a broader range of initial intensities is not explored. Third, the asymmetric response identified here is mainly derived from idealized model experiments and has not yet been quantitatively evaluated against observational TC–eddy interaction cases. It should therefore be interpreted primarily as a mechanistic demonstration rather than a direct quantitative constraint on real-world TC intensity changes. Further observational analysis and more realistic modeling are needed to assess the robustness and quantitative relevance of this asymmetry.

Overall, this study advances the understanding of ocean-TC interactions by quantifying an asymmetric intensity response to thermal forcing. These findings highlight that upper-ocean heat content and its spatial patchiness are critical predictors beyond sea surface temperature alone. This insight offers a pathway to improve the predictability of TC intensity, especially rapid changes, within evolving marine environments. Furthermore, this quantified nonlinear relationship provides a basis for developing more accurate parameterizations of ocean feedback in TC intensity forecast models.

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Author contributions
Conceptualization: TL. Methodology: WL, TL, JH and QX. Investigation: WL, TL, JH, QX and WS. Visualization: WL. Funding acquisition: TL. Supervision: TL and JH. Writing—original draft: WL. Writing—review & editing: TL, YG, CC, JJ and HZ.

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Data availability
Data used in this study are publicly available, with best-track records from https://www.ncdc.noaa.gov/ibtracs/, sea level anomaly data from the CMEMS (https://doi.org/10.48670/moi-00145), mesoscale eddy trajectories are taken from the altimetric Mesoscale Eddy Trajectories Atlas (META3.2 DT; https://www.aviso.altimetry.fr/en/data/products/value-added-products/global-mesoscale-eddy-trajectory-product/meta3-2-dt.html), atmospheric initial conditions from the ECMWF reanalysis dataset (https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels? tab=overview), and the ocean initial conditions from the CMEMS (https://doi.org/10.48670/moi-00021).

Declarations

Competing interest
The authors declare no competing interest.

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